ACADEMIC MAJOR PROJECT REPORT

TalentAI: Smart AI Job Finder & Career Growth Platform

An Intelligent Cross-Platform Mobile & Web Ecosystem Featuring Scam Job Detection, ATS Resume Optimization, and Real-Time Application Tracking

Project Domain: Artificial Intelligence & Full-Stack Web/Mobile	**Deployment:** Live Web (PWA) & Android Native APK
Target Audience: Engineering Students & Fresh Graduates	**Live URL:** https://talentai-rohan.surge.sh

1. Executive Summary & Abstract

TalentAI is an end-to-end career intelligence solution engineered to address critical inefficiencies in college placements and graduate hiring. Freshers face two monumental hurdles: falling prey to fraudulent job postings that charge fake fees, and having their resumes automatically filtered out by Applicant Tracking Systems (ATS). TalentAI combines an automated rule-based scam classifier, an instant ATS resume keyword optimizer with real-time PDF generation, a verified repository of 102+ active corporate opportunities (Google, Microsoft, Swiggy, Zomato, TCS), and an interactive Kanban application tracker into a single ultra-fast, mobile-friendly application.

2. Problem Statement & Motivation

- **Proliferation of Job Scams:** Fake recruiters exploit desperate job seekers by requiring 'security deposits', training fees, or redirecting to anonymous Telegram groups.
- **High ATS Rejection Rates (75%+):** Most student resumes lack the exact industry keywords or clean formatting required by modern automated hiring engines.
- **Disorganized Application Pipelines:** Students apply across 5+ disparate platforms (LinkedIn, Naukri, Indeed, Company portals) without an integrated milestone tracker.

3. Technical Architecture & System Modules

System Module	Engineering Implementation	Key Output
1. Verified Feed Aggregator	Ingests structured corporate catalog (102+ openings) with fallback to RapidAPI JSearch live web feeds.	Zero-latency instant rendering of top Indian tech jobs.
2. AI Scam Shield	Multi-rule heuristic analyzer scanning fee keywords, salary anomalies, and Telegram/WhatsApp contacts.	Visual red warning badges preventing candidate fraud.
3. ATS Resume Optimizer	Tokenizes candidate skill profile, matches against job requirements, and computes match % score.	Dynamic ATS PDF generation with live View & Download.
4. Kanban Tracker	Client-side pipeline management with isolated storage keys for each authenticated user.	Applied, Interviewing, Offer stages with celebration confetti.

4. Core Algorithms & Innovation

A. Automated Scam Detection Heuristics:

✔ **Financial Exactions:** Checks for tokens such as 'registration fee', 'deposit', 'training fee', 'processing fee'.

✔ **Communication Channels:** Flags non-corporate domain emails (@gmail.com) and unauthorized Telegram/WhatsApp channels.

✔ **Salary Outlier Check:** Cross-references experience requirements with compensation to flag absurd work-from-home claims.

✔ **Trust Badge Rendering:** High-risk listings are labeled with *'★ Suspicious Listing Risk'* and a detailed fraud explanation.

B. ATS Resume Matching & Scoring Formula:

✔ **Skill Vector Extraction:** Parses user skills (e.g. React, Node.js, Python, System Design) and compares with job prerequisites.

✔ **Jaccard Similarity / Weight:** *Score = (Candidate Skills ∩ Job Skills) / (Job Required Skills) * 100*.

✔ **Smart Resume Refactoring:** Injects missing industry keywords into project bullet points to maximize recruiter ATS ranking.

✔ **Instant PDF Compilation:** Generates a clean, single-column ATS-compliant document viewable directly in-app or downloadable.

5. Technology Stack & Specifications

Layer	Technologies Used	Justification
Frontend UI	HTML5, Tailwind CSS, Vanilla JS, Phosphor Icons	Sub-second initial paint, zero build overhead, responsive across mobile & desktop.
Mobile Runtime	Capacitor 8 (Android WebView), PWA Service Worker	Native Android APK deployment without rewriting codebase in Java/Kotlin.
Backend / Auth	Supabase Integration + Isolated LocalStorage	Supports cloud sync while guaranteeing complete client-side data privacy.
External APIs	RapidAPI JSearch + Direct Corporate ATS Portal Links	Connects real-time live web listings with 1-click external company apply links.

6. Teacher Viva / 2-Minute Summary Script

HOW TO EXPLAIN TO YOUR TEACHER IN 2 MINUTES:

*"Respected Teacher, our project is **TalentAI**. We developed it to tackle the huge problem of fake job scams and high ATS resume rejections that college students face during placements.*

Our application has 4 key innovations:
*1. **Verified Corporate Catalog:** We aggregated 102+ authentic tech jobs from top companies like Google, Swiggy, Zomato, TCS, and PhonePe with direct 1-click career portal links.*
*2. **AI Scam Detection Shield:** It automatically flags suspicious postings asking for registration fees or redirecting to Telegram, protecting students from fraud.*
*3. **ATS Resume Optimizer:** It calculates exact skill match percentages and automatically generates ATS-friendly PDF resumes that students can preview and download.*
*4. **Application Tracker:** A full Kanban tracking pipeline with celebratory confetti and milestone reminders.*

*We built this as a high-performance cross-platform system, packaged as an Android APK using Capacitor, and deployed live at **https://talentai-rohan.surge.sh**."*

7. Results & Future Roadmap

Current Status: TalentAI is 100% functional, responsive across mobile & desktop devices, verified with 102 active corporate roles, and deployed publicly for live demonstration.
Future Enhancements: (1) AI Mock Interview Bot with real-time audio analysis, (2) Direct recruiter Supabase chat dashboard, and (3) Automated email calendar integration for scheduled interview rounds.